

MULTI-OBJECTIVE COGNITIVE SCHEDULER: 6-TERM WORKED EVALUATION

$$\text{Score}(m) = F_{\text{cap}}(m, C_t) - \alpha M_{\text{cost}}(m) - \beta L_{\text{load}}(m) - \gamma E_{\text{cost}}(m) - \delta P_{\text{evict}}(m) + \eta F_{\text{fut}}(m, S_{\text{future}})$$

Terms: (1) Capability Fit - (2) RAM Footprint - (3) Load Stall - (4) Energy - (5) Evict Penalty + (6) Future Reuse

Current Workload State: Stage t demand C_t = Math Derivation | RAM Budget $B_{\text{RAM}} = 8.0 \text{ GB}$ | Upcoming Stage: Verification

Candidate 1: Qwen2.5-Math-7B

+ Capability Fit (F_{cap}):	+0.96
- Memory Cost (αM_{cost}):	-0.18
- Load Latency (βL_{load}):	-0.15
- Energy Draw (γE_{cost}):	-0.05
- Evict Penalty (δP_{evict}):	-0.02
+ Future Reuse (ηF_{fut}):	+0.32

Net Score: +0.88

DECISION: SELECTED (WINNER)

Candidate 2: Llama-3-70B

+ Capability Fit (F_{cap}):	+0.98
- Memory Cost (αM_{cost}):	-0.95
- Load Latency (βL_{load}):	-0.85
- Energy Draw (γE_{cost}):	-0.40
- Evict Penalty (δP_{evict}):	-0.30
+ Future Reuse (ηF_{fut}):	+0.28

Net Score: -1.24

DECISION: REJECTED (OOM / Stall)

Candidate 3: DeepSeek-Coder-6.7B

+ Capability Fit (F_{cap}):	+0.35
- Memory Cost (αM_{cost}):	-0.16
- Load Latency (βL_{load}):	-0.12
- Energy Draw (γE_{cost}):	-0.05
- Evict Penalty (δP_{evict}):	-0.02
+ Future Reuse (ηF_{fut}):	+0.12

Net Score: +0.12

DECISION: REJECTED (Fit Deficit)